

Autonomous Surface & Dive Robot for Underwater Depth Mapping

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Problem Statement

The goal of the autonomous underwater vehicle (AUV) is to map the underwater topography at Dana Point and illustrate changes in water temperature with depth to better understand the water dynamics.

Impact

- The topographic data can be used to predict sea currents, which are vital for understanding heat transfer in the ocean and how nutrients are distributed among species.
- Climate change has caused significant harm to the organisms in the ocean; we hope to better understand the underwater environment to protect ocean life.



Future Work

- Higher-density sensor sweeps would enhance the resolution of the topographic map, particularly in areas with complex seafloor geometry.
- Improving the AUV's depth and horizontal control would reduce drift-related noise in the data, making readings at specific coordinates more repeatable.

Autonomous Underwater Vehicle (AUV) Features

Electrical Schematic & Sensors

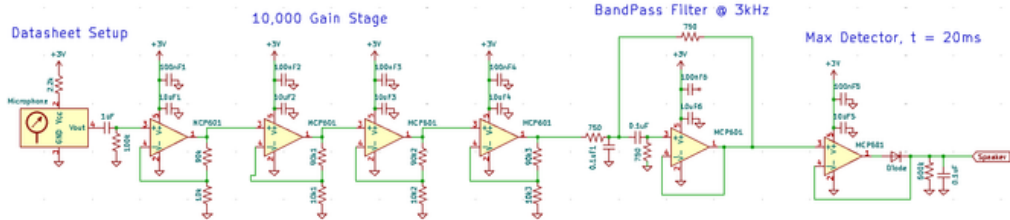
Temperature:

To analyze temperature measurements, the NXFT15WB473FA2B150, 47kOhm thermistor was used to create a voltage divider and had a wide range of temperatures (from 10°C to 19°C) at Dana Point Harbor.

Pressure:

For diving depth measurements, the MPX5700ASX absolute pressure sensor was used. The pressure was set to grow linearly downwards, with a depth range of 0 to 4.5 meters.

SONAR: For depth measurements, a CME-1538-100LB electret microphone and the daex25W-8 speaker were used to calculate the depth of the harbor floor. **TX and RX chains.**

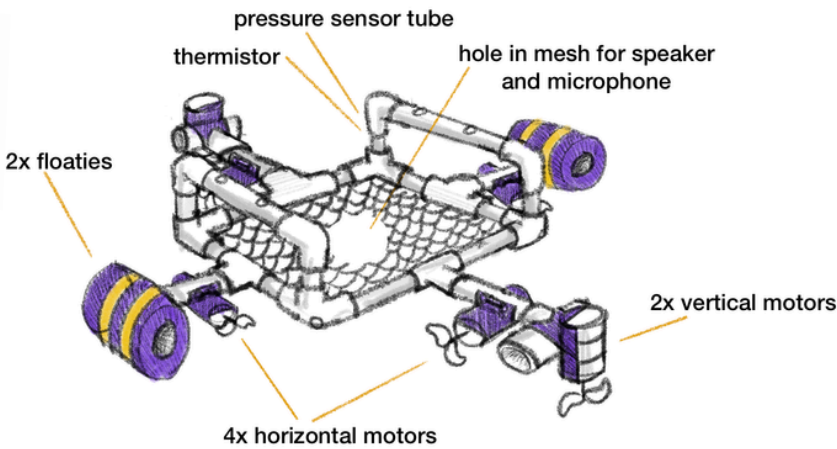


Mechanical Design & Integration

The design was chosen for its ability to dive quickly while also maintaining as much stability as possible on the surface.

Prominent features:

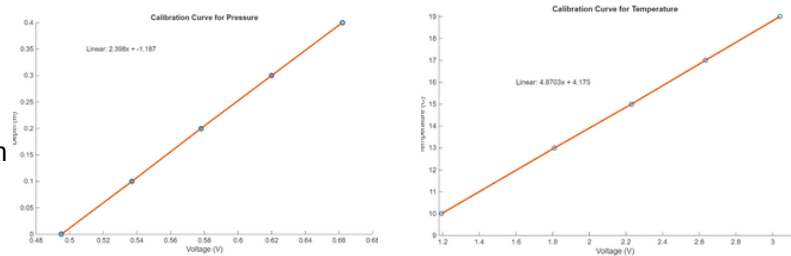
- Winch attached to fishing line for recovery
- Four horizontal and two vertical motors for increased ease of movement
- Low center of gravity compared to higher center of buoyancy



Experimental Data and Results

Sensor Calibration

- The pressure sensor was calibrated by measuring the output voltage at known depths. A graduated cylinder and meterstick were used.
- Five water baths were used to calibrate the thermistor. Measured the temperature of each bath and the output voltage of the circuit in each bath.
- To calibrate the sonar sensor, we took measurements at a set distance from the floor.

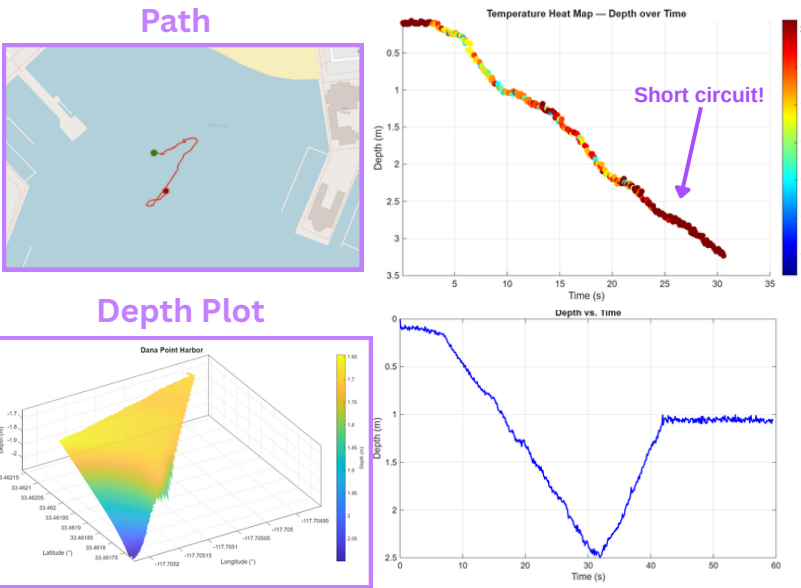


Ground Truth Measurements

- The team had two ground truth measurements:
- The temperature on the surface of the water was measured with a thermometer at the time of deployment at Dana Point
 - Depth at each point through the Dana Point website

Results & Conclusions

The chunk of the harbor floor was able to be mapped! For the dive portion, precise depth results by time were reported (temperature short-circuited unfortunately). Currents and other kayakers dragged our robot from its intended path, but the floor was mapped!



Deployment Details

- Location(s): Dana Point Harbor**
- Testing Protocol:** Either the surface or depth control was uploaded to the AUV. The team closed and secured the box onto the AUV frame, ensuring there were no wires that were caught. After each run, the team reads the data on the SD card to ensure it logged reasonable data.



Acknowledgements

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